

1. What is Natural Language Processing (NLP)? Mention any two real-world applications of NLP.
2. Describe the complete workflow of an NLP system from input text to output generation.
3. What is meant by an NLP pipeline? Explain its key phases.
4. How do syntactic analysis and semantic analysis contribute to language processing?
5. Define Computational Linguistics. How does it support NLP applications?
6. Discuss the significance of corpora in language processing tasks.
7. Describe the fundamental modules present in an NLP framework.
8. Why are syntax, semantics, and pragmatics considered essential levels of language analysis?
9. Explain the significance of contextual information in interpreting natural language.
10. Compare syntactic structure and semantic meaning with suitable examples.
11. What is TF-IDF weighting? Explain how it helps in identifying important terms in a document.
12. Explain the concept of Word2Vec embeddings and their usefulness in NLP tasks.
13. What are N-gram models? Illustrate their role in language prediction.
14. Discuss the application of statistical measures such as frequency count, mean, and variance in text analytics.
15. Explain the process of text normalization and its importance in NLP.
16. What is a corpus? Discuss its role in training and evaluating NLP systems.
17. Explain how Part-of-Speech (POS) tagging assigns grammatical categories to words.
18. Describe the purpose of lowercasing and character encoding in text preprocessing.
19. Differentiate between word-level and sentence-level tokenization with examples.
20. Why is text preprocessing considered a crucial step in NLP model development?
21. Illustrate the process of stopword elimination, stemming, and lemmatization using an example sentence.
22. Demonstrate how word tokenization, sentence tokenization, and subword tokenization are performed on a given text.
23. Explain the use of N-gram language models in predicting the next word of a sentence.
24. Analyze the importance of feature extraction techniques in machine learning-based NLP applications.
25. Explain how Probabilistic Context-Free Grammars (PCFGs) are used for parsing and assigning probabilities to sentence structures.
26. A language processing system needs to determine the most likely parse tree for a sentence.
 - a) Explain the concept of Probabilistic Context-Free Grammar (PCFG).
 - b) Demonstrate how rule probabilities are assigned and normalized in a PCFG.
 - c) Discuss the advantages of PCFG over ordinary CFG in syntactic parsing.
27. A company wants to automatically identify toxic comments from online discussions.
 - a) Design a toxicity detection framework using NLP techniques.
 - b) Explain the challenges involved in sarcasm detection and context understanding.

27. A media organization wants to identify names of politicians, countries, and institutions from news articles.
- a) Explain the concept of Named Entity Recognition .
 - b) Analyze the workflow of building an NER system.
 - c) Discuss the challenges of extracting entities from unstructured news text.
28. A virtual assistant should understand user queries in Hindi, English, and mixed-language text.
- a) Explain the role of POS tagging in multilingual NLP systems.
 - b) Analyze the challenges associated with code-mixed language processing.
 - c) Suggest suitable NLP models for multilingual intent classification.
29. A telecom company wants to convert customer speech into text and identify service requests.
- a) Explain the architecture of a speech recognition system.
 - b) Discuss the role of Hidden Markov Models (HMMs) in speech recognition.
 - c) Explain the challenges involved in recognizing spontaneous speech.
30. A healthcare provider wants to automatically extract symptoms, medicines, and diagnoses from clinical notes.
- a) Design an NLP-based information extraction system for healthcare records.
 - b) Explain the role of NER and POS tagging in extracting medical entities.
 - c) Discuss the challenges associated with processing clinical text data.
31. A university plans to deploy a chatbot to answer queries related to admissions, courses, and fees.
- a) Design an intent recognition and entity extraction framework for the chatbot.
 - b) Explain the NLP components required in the chatbot architecture.
 - c) Discuss the challenges of handling ambiguous and incomplete queries.
32. An e-commerce platform wants to generate summaries of customer reviews written in different languages.
- a) Design an NLP framework for multilingual text summarization.
 - b) Explain how word embeddings and Transformer models improve summarization quality.
 - c) Discuss challenges related to low-resource and code-mixed languages.
33. A legal consultancy wants to automatically identify obligations, deadlines, and parties involved in contracts.

- a) Design an NLP-based legal information extraction system.
- b) Explain the role of semantic analysis and NER in contract understanding.
- c) Discuss challenges associated with processing legal documents.

34. A university wants to classify student feedback into categories such as academics, facilities, and administration.

- a) Design a text classification framework using NLP techniques.
- b) Explain the importance of preprocessing and feature extraction in classification.
- c) Suggest methods for improving classification accuracy and robustness.

35. A smart learning platform wants to generate examination questions automatically from textbook content.

- a) Design an NLP-based question generation framework.
- b) Explain the role of POS tagging, parsing, and semantic analysis in question generation.

36. A travel assistant should understand commands such as “Reserve a hotel in Jaipur next weekend.”

- a) Design an intent recognition and slot-filling system for travel booking
- b) Explain the role of NER, semantic analysis, and dialogue management in understanding user requests.